Markov Chains and Stochastic Processes

A Comprehensive Analysis

Seminar Report

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Chapter 1

Introduction & DTMC Fundamentals

Markov chains are a powerful mathematical construct used to model a vast array of systems in computer science, statistics, physics, biology, operations research, and business. Their applications range from machine learning and system modeling (such as networking protocols, memory management, server performance, capacity provisioning, disk protocols) to supply chain and inventory management. Markov chains are pervasive—as you study them, look for their presence in your own work and the world around you; they are everywhere!

1.1 Definition[1]

A **Markov process** generalizes independent processes by allowing the outcome at any instant to depend only on the outcome that immediately precedes it—not the entire history. That is, in a Markov process, the future is independent of the past given the present. This is called the **Markov property** or **memorylessness**. Markov processes are named after Andrey A. Markov (1856–1922), who introduced them for discrete-parameter systems with finite state spaces. Extensions to countably infinite state spaces were pioneered by Kolmogorov and others.

Formally, for process $x(t)$ and times $t_{n-1} < t_n$, the Markov property states:

$$P[x(t_n) \leq x_n \mid x(t),\, t \leq t_{n-1}] = P[x(t_n) \leq x_n \mid x(t_{n-1})].$$

This means that, if you know the state at time t_{n-1}, then knowing the entire history upto t_{n-1} does not give you any more information , conditioned on the present, the future evolution does not depend on its historical path.

A special type of Markov process is a **Markov chain**, in which the system occupies a finite or countably infinite set of states $e_1, e_2, \ldots, e_j, \ldots$, and the future evolution depends only on the current state, not on the history. Markov chains can be discrete-time or continuous-time depending on whether indexing is discrete or continuous.

1.2 Discrete-Time vs Continuous-Time Markov Chains[2]

Discrete-Time Markov Chains (DTMCs): DTMCs operate in synchronized time steps. Events such as arrivals or departures occur only at the end of each time step. This makes DTMCs somewhat artificial for some computer system models but suitable for many other problems.
Continuous-Time Markov Chains (CTMCs): CTMCs allow events to happen at any instant, providing a more natural model for systems with continuous evolution.

1.3 State Spaces (Finite vs Infinite)

A **state space** is the set of all possible states a Markov chain can occupy. The state space can be either:

- **Finite:** The number of possible states is limited and countable, e.g., $\{0, 1, 2, \ldots, M-1\}$. Analysis involves finite-dimensional transition matrices and is typically easier. Systems like board games, weather with a fixed set of conditions, or bounded random walks use finite state spaces.
- **Infinite:** The number of possible states is countably infinite or unbounded (e.g., all non-negative integers). Infinite-state Markov chains appear in queueing models or population dynamics where the number of jobs or entities can grow without bound. Their transition matrices are infinite-dimensional, and additional mathematical care is needed to analyze long-term behavior and existence of stationary distributions.

1.4 Transition Matrices[1]

The **transition probability matrix** associated with any DTMC is a matrix $\mathbf{P}$ whose (i, j)th entry, P_{ij}, represents the probability of moving to state j on the next transition, given that the current state is i.

Definition 1.1 (Transition Probability Matrix)**.** The transition probability matrix $\mathbf{P}$ for a Markov chain is defined as

$$\mathbf{P} = \begin{pmatrix} P_{11} & P_{12} & P_{13} & \cdots & P_{1J} \\ P_{21} & P_{22} & P_{23} & \cdots & P_{2J} \\ P_{31} & P_{32} & P_{33} & \cdots & P_{3J} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ P_{I1} & P_{I2} & P_{I3} & \cdots & P_{IJ} \end{pmatrix}$$

where $P_{ij} = P\{X_{n+1} = e_j \mid X_n = e_i\}$ is the probability of transition from state i to state j in one step.

Each row of $\mathbf{P}$ sums to one, reflecting the property that, given the chain is in state i, it must transition to one of the possible states:

$$\sum_j P_{ij} = 1 \qquad \forall i$$

When the Markov chain has infinitely many states, $\mathbf{P}$ may be of infinite order. By definition, the transition matrix is **stochastic**: all entries are non-negative, and every row sums to unity.

For homogeneous Markov chains, the transition probabilities are stationary (independent of time), and $\mathbf{P}$ completely determines the evolution of the chain alongside the initial distribution:

$$P_k(0) \triangleq P(X_0 = e_k)$$

1.5 Basic Examples

1.5.1 Weather Model Markov Chain

Consider a simplified weather system with two states: *Sunny* and *Rainy*. The weather evolves daily according to the following rules:

- If today is Sunny, tomorrow will be Sunny with probability 0.9 and Rainy with probability 0.1.
- If today is Rainy, tomorrow will be Rainy with probability 0.5 and Sunny with probability 0.5.

The state space is

$$S = \{\text{Sunny}, \text{Rainy}\}.$$

The transition probability matrix $\mathbf{P}$ is given by

$$\mathbf{P} = \begin{bmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{bmatrix},$$

where the rows represent the current state, and the columns represent the next day's weather state.

This model satisfies the **Markov property**: the future weather depends only on the current weather, not the entire history. Each row of $\mathbf{P}$ sums to 1, as they are proper probability distributions over next states.

By applying repeated powers of $\mathbf{P}$, one can compute the likelihood of weather on any future day. Additionally, the model may converge to a *stationary distribution* indicating the long-term frequency of Sunny and Rainy days.

This example demonstrates the key aspects of Markov chains: a finite state space, a transition probability matrix, and the memoryless property.

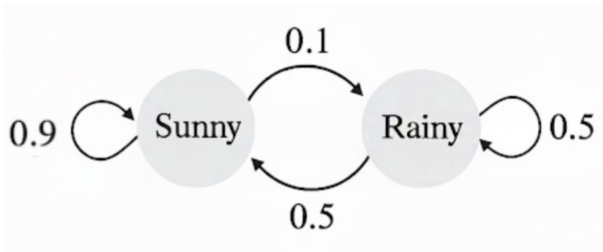

Figure 1.1: Markov chain for weather problem

1.5.2 Umbrella Problem

An absent-minded student owns two umbrellas and commutes daily between home and the college. Each time she commutes, if it rains and there is an umbrella at her current location, she takes one. If it is not raining, she always forgets to bring an umbrella. Suppose that it rains independently with probability p on any commute.

State Space: The state at any time is determined by how many umbrellas are at the **current** location (regardless of whether it is home or office). Thus, the Markov chain can be modeled with **three states**:

$$S_0 : \text{No umbrellas at the current location}$$
$$S_1 : \text{One umbrella at the current location}$$
$$S_2 : \text{Two umbrellas at the current location}$$

Transition Probability Matrix: Let the states be ordered as $0, 1, 2$ (umbrellas at the current location). The transition matrix is

$$\mathbf{P} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1-p & p \\ 1-p & p & 0 \end{bmatrix}$$

where entry (i, j) gives the probability of moving from state i to state j in one step.

- When in state 2 (both umbrellas at the current location), and it rains, she takes one, moving to state 1 at the next commute; if it does not rain, she forgets, and the number remains unchanged.
- Similarly, interpret transitions for other cases by considering the "forgetting" rule when it does not rain.

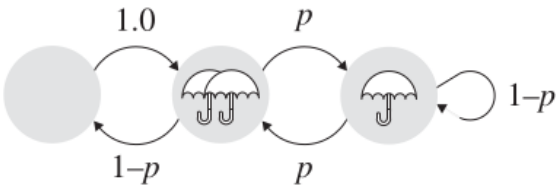

Figure 1.2: DTMC state diagram for the umbrella problem.

Chapter 2

Chapman-Kolmogorov Equations & Multi-Step Analysis

2.1 Chapman-Kolmogorov Equation: Statement and Derivation[3]

The **Chapman-Kolmogorov equation** describes multi-step transitions in Markov chains. It states that the probability of transitioning from state i to state j in $m+n$ steps is given by

$$P^{(m+n)}(i,j) = \sum_k P^{(m)}(i,k)P^{(n)}(k,j),$$

where

- $P^{(m+n)}(i,j)$ is the probability of transitioning from state i to state j in $m+n$ steps,
- the summation is over all intermediate states k.

Derivation

The derivation starts from the law of total probability. For a discrete-time Markov chain,

$$P(X_{m+n} = j \mid X_0 = i) = \sum_k P(X_{m+n} = j, X_m = k \mid X_0 = i).$$

Using the definition of conditional probability and the Markov property,

$$P(X_{m+n} = j \mid X_0 = i) = \sum_k P(X_{m+n} = j \mid X_m = k, X_0 = i) \cdot P(X_m = k \mid X_0 = i).$$

By the Markov property, the future state depends only on the present state, thus

$$P(X_{m+n} = j \mid X_m = k, X_0 = i) = P(X_n = j \mid X_0 = k).$$

Therefore,

$$P(X_{m+n} = j \mid X_0 = i) = \sum_k P(X_m = k \mid X_0 = i) \cdot P(X_n = j \mid X_0 = k),$$

or equivalently,

$$P^{(m+n)}(i,j) = \sum_k P^{(m)}(i,k)P^{(n)}(k,j).$$

Matrix Notation

For a finite state space, this equation is equivalent to the matrix multiplication

$$\mathbf{P}^{(m+n)} = \mathbf{P}^{(m)}\mathbf{P}^{(n)},$$

and for a time-homogeneous Markov chain,

$$\mathbf{P}^{(n)} = \mathbf{P}^{n}.$$

2.2 Matrix Powers and Computational Aspects[1]

Computing $\mathbf{P}^n$ directly by matrix multiplication is costly; complexity is approximately $O(N^3 n)$ for an $N \times N$ matrix.

An efficient method uses eigenvalue decomposition. For diagonalizable $\mathbf{P}$:

$$\mathbf{P} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^{-1},$$

where $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_N)$ contains eigenvalues.

Matrix powers become:

$$\mathbf{P}^n = \mathbf{U}\boldsymbol{\Lambda}^n\mathbf{U}^{-1} = \mathbf{U}\begin{pmatrix} \lambda_1^n & & \\ & \ddots & \\ & & \lambda_N^n \end{pmatrix}\mathbf{U}^{-1}.$$

This reduces complexity substantially and also provides insight into long-term behavior of the chain.

2.3 Higher-order Transition Behavior[1]

For irreducible, aperiodic Markov chains, the n-step distributions converge to a stationary distribution π:

$$\lim_{n\to\infty} P_{ij}^{(n)} = \pi_j.$$

Chapter 3

Classification of States Theory

3.1 Communication and Accessibility[4]

State j is **accessible** from state i if there exists a positive integer n such that the n-step transition probability $P_{ij}^{(n)} > 0$. This means you can reach j starting from i with positive probability.

Two states i and j **communicate** if each is accessible from the other ($i \leftrightarrow j$). The communication relation is an **equivalence relation** because it is reflective, symmetric, and transitive, partitioning the state space into *communicating classes*—subsets within which all states communicate with each other.

3.2 Communicating Classes and Irreducibility[5]

A **communicating class** is a set of states that all communicate with each other and are disjoint from other classes. A class is **closed** if no state outside it is accessible from any state inside the class. If a Markov chain consists of only one communicating class, it is called **irreducible**—meaning every state can be reached from every other state.

An **absorbing state** is a special state where once entered, the process cannot leave ($P_{ii} = 1$). A Markov chain containing at least one absorbing state is called an *absorbing Markov chain*.

3.3 Transient vs Recurrent State Definitions[6]

Define the **return probability** f_{ii} as the probability of eventually returning to state i starting from i. Then a state is:

- **Recurrent** if $f_{ii} = 1$, meaning the chain almost surely returns to that state infinitely often.
- **Transient** if $f_{ii} < 1$, so there is some positive probability the chain never returns.

Recurrence and transience are class properties: if two states communicate, then they are either both recurrent or both transient.

Recurrent states are further classified as:

- **Positive recurrent** if the expected return time is finite.
- **Null recurrent** if the expected return time is infinite.

Finite irreducible Markov chains have all states positive recurrent.

3.4 Periodicity and Aperiodicity[7]

The **period** of a state i, denoted $d(i)$, is the greatest common divisor of the set of n such that $P_{ii}^{(n)} > 0$. If $d(i) > 1$, the state is *periodic*; if $d(i) = 1$, it is *aperiodic*.

Periodicity is also a class property; all states in a communicating class share the same period. An irreducible chain is aperiodic if all its states are aperiodic.

Chapter 4

Stationary Distributions & Existence Theory[1][2]

In the study of Markov chains, one of the most fundamental questions concerns their long-term behavior: Does the chain settle into a stable pattern over time? This chapter explores the theoretical foundations of stationary distributions, which capture the essence of such stable long-term behavior in Markov chains.

4.1 Definition of Stationary Distributions

A stationary distribution represents a probability distribution over the states of a Markov chain that remains unchanged as the chain evolves. This concept is central to understanding the equilibrium behavior of stochastic processes.

Definition 4.1 (Stationary Distribution)**.** For a discrete-time Markov chain with transition probability matrix $\mathbf{P}$ and state space S, a probability distribution $\boldsymbol{\pi} = (\pi_1, \pi_2, \ldots)$ is called a **stationary distribution** if it satisfies:

1. $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$ (the stationary equation)
2. $\sum_{i \in S} \pi_i = 1$ (normalization condition)
3. $\pi_i \geq 0$ for all $i \in S$ (non-negativity)

The stationary equation $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$ can be written component-wise as:

$$\pi_j = \sum_{i \in S} \pi_i P_{ij} \quad \text{for all } j \in S$$

This equation states that if the chain is distributed according to $\boldsymbol{\pi}$ at some time n, then it will also be distributed according to $\boldsymbol{\pi}$ at time $n + 1$. Hence the name "stationary."

4.1.1 Interpretation of Stationary Distributions

The stationary distribution has several important interpretations:

1. **Equilibrium Distribution**: If a Markov chain starts with the stationary distribution, it remains in that distribution forever.

2. **Long-run Proportion**: For ergodic chains, π_j represents the long-run proportion of time the chain spends in state j.

3. **Limiting Distribution**: Under certain conditions, $\pi_j = \lim_{n\to\infty} P_{ij}^{(n)}$ regardless of the starting state i.

4.2 Existence and Uniqueness Theorems

The existence and uniqueness of stationary distributions depend on the structural properties of the Markov chain. The key results are summarized in the following fundamental theorems.

Theorem 4.1 (Existence for Finite Chains)**.** Every finite Markov chain has at least one stationary distribution.

Proof. For a finite chain with transition matrix $\mathbf{P}$, consider the eigenvalue equation $\boldsymbol{\pi}\mathbf{P} = \lambda\boldsymbol{\pi}$. Since $\mathbf{P}$ is stochastic, we have $\mathbf{P}\mathbf{1} = \mathbf{1}$, which implies that 1 is an eigenvalue of $\mathbf{P}$. Therefore, there exists a left eigenvector corresponding to eigenvalue 1. By normalizing this eigenvector to sum to 1 and taking absolute values to ensure non-negativity, we obtain a stationary distribution. □

Theorem 4.2 (Fundamental Theorem for Irreducible Chains)**.** Let $\{X_n\}$ be an irreducible discrete-time Markov chain. Then:

1. If the chain is positive recurrent, then there exists a unique stationary distribution $\boldsymbol{\pi}$ with $\pi_i > 0$ for all states i.

2. If the chain is null recurrent or transient, then no stationary distribution exists.

This theorem establishes the fundamental connection between the recurrence properties of states and the existence of stationary distributions.

Theorem 4.3 (Uniqueness for Finite Irreducible Chains)**.** A finite irreducible Markov chain has a unique stationary distribution $\boldsymbol{\pi}$ with $\pi_i > 0$ for all states i.

Proof. Since the chain is finite and irreducible, all states are positive recurrent. By the fundamental theorem above, a unique stationary distribution exists. The positivity follows from irreducibility: if $\pi_i = 0$ for some state i, then the stationary equation would imply that the chain never enters state i, contradicting irreducibility. □

4.2.1 Perron-Frobenius Theory

For primitive matrices (irreducible and aperiodic), the Perron-Frobenius theorem provides deeper insights:

Theorem 4.4 (Perron-Frobenius for Stochastic Matrices)**.** Let $\mathbf{P}$ be a primitive stochastic matrix. Then:

1. The eigenvalue 1 has algebraic and geometric multiplicity 1

2. All other eigenvalues have absolute value strictly less than 1

3. There exists a unique stationary distribution $\boldsymbol{\pi}$ with all positive entries

4. $\lim_{n\to\infty} \mathbf{P}^n = \mathbf{1}\boldsymbol{\pi}^T$

This theorem guarantees convergence to the stationary distribution from any initial distribution.

4.3 Methods for Computing Stationary Distributions

Several computational approaches exist for finding stationary distributions, each with its own advantages and limitations.

4.3.1 Direct Solution of Linear System

The most straightforward approach involves solving the linear system:

$$\begin{cases} \boldsymbol{\pi}(\mathbf{P} - \mathbf{I}) = \mathbf{0} \\ \sum_i \pi_i = 1 \end{cases}$$

This can be rewritten as:

$$\begin{cases} \boldsymbol{\pi}(\mathbf{I} - \mathbf{P})^T = \mathbf{0} \\ \boldsymbol{\pi}\mathbf{1} = 1 \end{cases}$$

For computational purposes, we replace one equation with the normalization constraint to ensure a unique solution.

4.3.2 Power Method

The power method exploits the convergence property $\mathbf{P}^n \to \mathbf{1}\boldsymbol{\pi}^T$:

1. Start with an arbitrary probability distribution $\boldsymbol{\mu}^{(0)}$
2. Iterate: $\boldsymbol{\mu}^{(n+1)} = \boldsymbol{\mu}^{(n)}\mathbf{P}$
3. Continue until convergence: $\boldsymbol{\mu}^{(n)} \to \boldsymbol{\pi}$

The convergence rate depends on the second-largest eigenvalue in absolute value.

4.3.3 Eigenvector Methods

For the eigenvalue formulation $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$, we can:

1. Compute the left eigenvector of $\mathbf{P}$ corresponding to eigenvalue 1
2. Use numerical eigenvalue algorithms (QR, Arnoldi, etc.)
3. For large sparse matrices, use iterative methods like the Arnoldi iteration

4.3.4 Monte Carlo Methods

For very large state spaces, simulation-based approaches become necessary:

1. **Markov Chain Monte Carlo (MCMC)**: Run long simulations and estimate π_i as the empirical frequency of visits to state i
2. **Coupling from the past**: Ensures exact sampling from the stationary distribution

4.3.5 Computational Complexity

The choice of method depends on the problem size and structure:

- Direct solution: $O(n^3)$ for dense matrices
- Power method: $O(kn^2)$ where k is the number of iterations
- Sparse eigenvalue methods: $O(kn)$ for sparse matrices
- Monte Carlo: Depends on mixing time and desired accuracy

4.4 Balance Equations and Their Interpretation

The stationary equations can be interpreted through the lens of probability flow, leading to powerful insights about equilibrium behavior.

4.4.1 Global Balance Equations

The stationary equation $\pi_j = \sum_i \pi_i P_{ij}$ is called the **global balance equation** for state j. It can be rewritten as:

$$\pi_j \sum_{k \neq j} P_{jk} = \sum_{i \neq j} \pi_i P_{ij}$$

This form reveals the balance interpretation:

- **Left side**: Rate at which probability flows out of state j
- **Right side**: Rate at which probability flows into state j

4.4.2 Flow Interpretation

Consider a large number of independent copies of the Markov chain, all evolving according to the same transition probabilities. In the stationary regime:

- π_i represents the fraction of chains currently in state i
- $\pi_i P_{ij}$ represents the flow rate from state i to state j
- The global balance equation ensures that flow in equals flow out for each state

4.4.3 Detailed Balance

A stronger condition than global balance is **detailed balance**:

Definition 4.2 (Detailed Balance)**.** A Markov chain satisfies detailed balance if:

$$\pi_i P_{ij} = \pi_j P_{ji} \quad \text{for all states } i, j$$

Detailed balance implies that the flow from i to j exactly equals the flow from j to i for every pair of states.

Theorem 4.5 (Detailed Balance Implies Global Balance)**.** If a distribution $\boldsymbol{\pi}$ satisfies detailed balance, then it satisfies global balance and is therefore a stationary distribution.

Proof. Summing the detailed balance equations over all i:

$$\sum_i \pi_i P_{ij} = \sum_i \pi_j P_{ji} = \pi_j \sum_i P_{ji} = \pi_j$$

The last equality uses the fact that $\sum_i P_{ji} = 1$. □

4.4.4 Reversibility

Chains satisfying detailed balance are called **reversible** because the process looks statistically identical when run forward or backward in time.

Theorem 4.6 (Characterization of Reversibility)**.** An irreducible Markov chain is reversible if and only if for every sequence of states $i_0, i_1, \ldots, i_k$ with $i_0 = i_k$:

$$P_{i_0 i_1} P_{i_1 i_2} \cdots P_{i_{k-1} i_k} = P_{i_k i_{k-1}} P_{i_{k-1} i_{k-2}} \cdots P_{i_1 i_0}$$

This provides a practical test for reversibility without computing the stationary distribution.

4.5 Examples with Detailed Calculations

We now work through several detailed examples to illustrate the computation of stationary distributions and the application of balance equations.

4.5.1 Two-State Chain

Consider a two-state Markov chain with states $\{0, 1\}$ and transition matrix:

$$\mathbf{P} = \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix}$$

where $0 < \alpha, \beta < 1$.

Method 1: Direct Solution

The stationary equations are:

$$\pi_0 = (1-\alpha)\pi_0 + \beta\pi_1 \tag{4.1}$$

$$\pi_1 = \alpha\pi_0 + (1-\beta)\pi_1 \tag{4.2}$$

$$\pi_0 + \pi_1 = 1 \tag{4.3}$$

From the first equation:

$$\pi_0 = (1-\alpha)\pi_0 + \beta\pi_1 \implies \alpha\pi_0 = \beta\pi_1 \implies \pi_1 = \frac{\alpha}{\beta}\pi_0$$

Substituting into the normalization condition:

$$\pi_0 + \frac{\alpha}{\beta}\pi_0 = 1 \implies \pi_0\left(1 + \frac{\alpha}{\beta}\right) = 1 \implies \pi_0 = \frac{\beta}{\alpha+\beta}$$

Therefore:

$$\boldsymbol{\pi} = \left(\frac{\beta}{\alpha+\beta}, \frac{\alpha}{\alpha+\beta}\right)$$

Method 2: Detailed Balance
Assume detailed balance: $\pi_0 P_{01} = \pi_1 P_{10}$

$$\pi_0 \alpha = \pi_1 \beta \implies \frac{\pi_1}{\pi_0} = \frac{\alpha}{\beta}$$

With normalization $\pi_0 + \pi_1 = 1$, we get the same result.
Verification:

$$\frac{\beta}{\alpha+\beta} \cdot \alpha = \frac{\alpha\beta}{\alpha+\beta} = \frac{\alpha}{\alpha+\beta} \cdot \beta$$

4.5.2 Birth-Death Process

Consider a birth-death process with states $\{0, 1, 2, \ldots\}$ where:

- Birth rate from state i: λ_i (transition to state $i+1$)
- Death rate from state i: μ_i (transition to state $i-1$)
- $P_{i,i+1} = \lambda_i/(\lambda_i + \mu_i)$, $P_{i,i-1} = \mu_i/(\lambda_i + \mu_i)$ for $i \geq 1$
- $P_{0,1} = 1$ (pure birth from state 0)

Detailed Balance Approach:
For $i \geq 1$:

$$\pi_i \frac{\lambda_i}{\lambda_i + \mu_i} = \pi_{i+1} \frac{\mu_{i+1}}{\lambda_{i+1} + \mu_{i+1}}$$

This simplifies to:

$$\pi_i \lambda_i = \pi_{i+1} \mu_{i+1}$$

Solving recursively:

$$\pi_1 = \frac{\lambda_0}{\mu_1}\pi_0, \quad \pi_2 = \frac{\lambda_1}{\mu_2}\pi_1 = \frac{\lambda_0\lambda_1}{\mu_1\mu_2}\pi_0$$

In general:

$$\pi_n = \frac{\lambda_0\lambda_1\cdots\lambda_{n-1}}{\mu_1\mu_2\cdots\mu_n}\pi_0 = \pi_0 \prod_{k=0}^{n-1} \frac{\lambda_k}{\mu_{k+1}}$$

Existence Condition:
For the stationary distribution to exist, we need:

$$\sum_{n=0}^{\infty} \pi_n = \pi_0 \left(1 + \sum_{n=1}^{\infty} \prod_{k=0}^{n-1} \frac{\lambda_k}{\mu_{k+1}}\right) = 1$$

This requires convergence of the series $\sum_{n=1}^{\infty} \prod_{k=0}^{n-1} \frac{\lambda_k}{\mu_{k+1}}$.
Special Case: M/M/1 Queue
For constant rates $\lambda_i = \lambda$ and $\mu_i = \mu$ (for $i \geq 1$):

$$\pi_n = \left(\frac{\lambda}{\mu}\right)^n \pi_0 = \rho^n \pi_0$$

where $\rho = \lambda/\mu$ is the traffic intensity.
For convergence, we need $\rho < 1$, giving:

$$\pi_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \rho^n} = \frac{1}{1 + \rho/(1-\rho)} = 1 - \rho$$

Therefore:

$$\pi_n = (1-\rho)\rho^n \quad \text{for } n \geq 0$$

This is the well-known geometric distribution for the M/M/1 queue.

4.5.3 Ehrenfest Model

The Ehrenfest model represents the exchange of molecules between two containers. With N total molecules and states $\{0, 1, 2, \ldots, N\}$ representing the number of molecules in container 1:

$$P_{i,i+1} = \frac{N-i}{N}, \quad P_{i,i-1} = \frac{i}{N}$$

Detailed Balance:

$$\pi_i \frac{N-i}{N} = \pi_{i+1} \frac{i+1}{N}$$

This gives:

$$\pi_{i+1} = \frac{N-i}{i+1}\pi_i$$

Solving recursively:

$$\pi_i = \frac{N(N-1)\cdots(N-i+1)}{i!}\pi_0 = \binom{N}{i}\pi_0$$

Using normalization:

$$\sum_{i=0}^{N} \pi_i = \pi_0 \sum_{i=0}^{N} \binom{N}{i} = \pi_0 \cdot 2^N = 1$$

Therefore:

$$\pi_i = \frac{1}{2^N}\binom{N}{i}$$

This is the binomial distribution, reflecting the fact that each molecule chooses its container independently with probability 1/2.

4.6 Conclusion

Stationary distributions provide fundamental insights into the long-term behavior of Markov chains. The existence and uniqueness theorems establish when such distributions exist, while various computational methods enable their calculation. The balance equation interpretation offers intuitive understanding through probability flow concepts, and detailed balance provides a powerful tool for analysis and computation.

The examples demonstrate the practical application of these concepts across diverse domains, from simple two-state systems to complex queueing models. Understanding stationary distributions is essential for analyzing the equilibrium behavior of stochastic systems and forms the foundation for more advanced topics in Markov chain theory.

Chapter 5

Ergodicity Theory

Ergodicity theory is a fundamental branch of mathematics that studies the statistical properties of deterministic dynamical systems and stochastic processes. It provides the theoretical foundation for connecting time averages with ensemble averages, which is crucial in probability theory, statistical mechanics, and the analysis of Markov chains.

5.1 Ergodic Theorem for Markov Chains[14]

The Ergodic Theorem for Markov chains is a fundamental result that establishes conditions under which time averages converge to their expected values. This theorem is particularly important for understanding the long-term behavior of Markov chains.

Theorem 5.1 (Ergodic Theorem for Markov Chains)**.** Let $(X_n)_{n\geq 0}$ be an irreducible and positive recurrent Markov chain with state space $\mathcal{S}$ and stationary distribution π. For any state $i \in \mathcal{S}$ and any initial distribution, we have:

$$\lim_{n\to\infty} \frac{V_i(n)}{n} = \pi_i$$

with probability 1, where $V_i(n) = \sum_{k=0}^{n-1} \mathbf{1}_{\{X_k=i\}}$ is the number of visits to state i before time n.

The theorem states that the long-run proportion of time spent in state i equals the stationary probability π_i. This provides a bridge between the microscopic behavior of the chain (individual transitions) and its macroscopic properties (stationary distribution).

Corollary 5.1 (Fundamental Limit Theorem)**.** For an irreducible, aperiodic, and positive recurrent Markov chain:

$$\lim_{n\to\infty} P_{ij}^{(n)} = \pi_j$$

for all states i, j, where $P_{ij}^{(n)}$ is the n-step transition probability from state i to state j.

5.1.1 Conditions for Ergodicity

A Markov chain is called **ergodic** if it satisfies:

1. **Irreducibility**: Every state can be reached from every other state with positive probability
2. **Aperiodicity**: The chain does not exhibit periodic behavior
3. **Positive recurrence**: All states have finite expected return times

Theorem 5.2 (Fundamental Theorem of Markov Chains)**.** A finite-state Markov chain is ergodic if and only if it is irreducible and aperiodic. Such chains have:

1. A unique stationary distribution π with $\pi_i > 0$ for all states i
2. Convergence: $\lim_{n\to\infty} P_{ij}^{(n)} = \pi_j$ for all i, j
3. The ergodic property: time averages equal ensemble averages with probability 1

5.2 Time Averages vs Ensemble Averages[15]

The distinction between time averages and ensemble averages is central to ergodic theory and statistical mechanics. Understanding this relationship is crucial for interpreting the behavior of stochastic systems.

5.2.1 Definitions

Definition 5.1 (Time Average)**.** For a stochastic process $(X_t)_{t\geq 0}$ or sequence $(X_n)_{n\geq 0}$, the time average of a function f is:

$$\langle f\rangle_{\text{time}} = \lim_{T\to\infty} \frac{1}{T}\int_0^T f(X_t)dt$$

or in discrete time:

$$\langle f\rangle_{\text{time}} = \lim_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1} f(X_k)$$

Definition 5.2 (Ensemble Average)**.** The ensemble average (or expectation) of a function f is:

$$\langle f\rangle_{\text{ensemble}} = \mathbb{E}[f(X)] = \sum_x f(x)\pi(x)$$

where π is the stationary distribution.

5.2.2 The Ergodic Hypothesis

The **ergodic hypothesis** states that for ergodic systems:

$$\langle f\rangle_{\text{time}} = \langle f\rangle_{\text{ensemble}}$$

with probability 1.

This equality is fundamental because:

- Time averages can be computed from a single long trajectory
- Ensemble averages require knowledge of the stationary distribution
- The ergodic theorem guarantees their equivalence under appropriate conditions

5.2.3 Physical Interpretation

In statistical mechanics, this relationship means:

- **Time average**: Observing a single system over a long period
- **Ensemble average**: Taking simultaneous measurements across many identical systems
- **Ergodicity**: Both approaches yield the same result

5.3 Mean Return Times and Their Calculation[16]

Mean return times are fundamental quantities in Markov chain theory that characterize how quickly a chain returns to specific states.

Definition 5.3 (Mean Return Time)**.** For a Markov chain and state i, the mean return time is:

$$m_i = \mathbb{E}_i[T_i]$$

where $T_i = \inf\{n \geq 1 : X_n = i\}$ is the first return time to state i starting from state i.

5.3.1 Fundamental Relationship

Theorem 5.3 (Mean Return Time Formula)**.** For an irreducible, positive recurrent Markov chain with stationary distribution π:

$$m_i = \frac{1}{\pi_i}$$

This remarkable result shows that states with higher stationary probabilities have shorter expected return times, which aligns with intuition.

5.3.2 Calculation Methods

Method 1: First-Step Analysis

Using the law of total expectation:

$$m_i = 1 + \sum_{j \neq i} p_{ij} \mathbb{E}_j[T_i]$$

where $\mathbb{E}_j[T_i]$ is the expected hitting time from state j to state i.

Method 2: System of Linear Equations

For hitting times $h_{ji} = \mathbb{E}_j[T_i]$:

$$h_{ii} = 0 \tag{5.1}$$

$$h_{ji} = 1 + \sum_{k \neq i} p_{jk} h_{ki} \quad \text{for } j \neq i \tag{5.2}$$

Then $m_i = 1 + \sum_{j \neq i} p_{ij} h_{ji}$.

Method 3: Using Stationary Distribution

Once the stationary distribution π is known:

$$m_i = \frac{1}{\pi_i}$$

Example 5.1 (Two-State Markov Chain)**.** Consider a chain with transition matrix:

$$P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$$

The stationary distribution is $\pi = \left(\frac{b}{a+b}, \frac{a}{a+b}\right)$.

Therefore:

$$m_1 = \frac{1}{\pi_1} = \frac{a+b}{b} \tag{5.3}$$

$$m_2 = \frac{1}{\pi_2} = \frac{a+b}{a} \tag{5.4}$$

5.4 Positive Recurrent vs Null Recurrent States[17]

The classification of recurrent states into positive and null recurrent is crucial for understanding the long-term behavior of Markov chains.

5.4.1 Basic Classifications

Definition 5.4 (State Classifications)**.** For a Markov chain, a state i is:

- **Transient** if $\mathbb{P}_i(T_i < \infty) < 1$
- **Recurrent** if $\mathbb{P}_i(T_i < \infty) = 1$
- **Positive recurrent** if recurrent and $\mathbb{E}_i[T_i] < \infty$
- **Null recurrent** if recurrent and $\mathbb{E}_i[T_i] = \infty$

5.4.2 Key Properties

Theorem 5.4 (Classification Properties)**.**

1. Recurrence and transience are class properties: if states communicate, they have the same classification
2. In finite state spaces, all recurrent states are positive recurrent
3. For infinite state spaces, both positive and null recurrence are possible

5.4.3 Relationship to Stationary Distributions

Theorem 5.5 (Stationary Distribution Existence)**.** For an irreducible Markov chain:

- A stationary distribution exists if and only if all states are positive recurrent
- If all states are null recurrent or transient, no stationary distribution exists
- When it exists, the stationary distribution is unique and given by $\pi_i = \frac{1}{m_i}$

5.4.4 Examples

Example 5.2 (Simple Random Walk)**.** Consider the simple symmetric random walk on $\mathbb{Z}$:

- In dimensions 1 and 2: null recurrent
- In dimensions 3 and higher: transient
- No stationary distribution exists (infinite state space, null recurrent/transient)

Example 5.3 (Birth-Death Chain)**.** For a discrete-time birth-death chain with birth probability $1-p$ and death probability p:

- If $p < 1/2$: state 0 is transient
- If $p > 1/2$: state 0 is positive recurrent
- If $p = 1/2$: state 0 is null recurrent

5.5 Long-run Proportion Interpretations[18]

The long-run proportion interpretation provides practical meaning to the theoretical results of ergodic theory and connects abstract mathematical concepts to observable phenomena.

5.5.1 The Fundamental Interpretation

For an ergodic Markov chain, the stationary probability π_i has several equivalent interpretations:

Theorem 5.6 (Long-run Proportion Interpretations)**.** For an irreducible, positive recurrent Markov chain, the following are equal:

1. π_i (stationary probability)
2. $\lim_{n\to\infty} \frac{V_i(n)}{n}$ (long-run proportion of time in state i)
3. $\lim_{n\to\infty} P_{ji}^{(n)}$ for any j (limiting probability)
4. $\frac{1}{m_i}$ (reciprocal of mean return time)

5.5.2 Practical Applications

Frequency Interpretation

$$\pi_i = \text{long-run frequency of visits to state } i$$

This means that in a long sequence of observations, approximately a fraction π_i of the time will be spent in state i.

Steady-State Interpretation

$$\lim_{n\to\infty} \mathbb{P}(X_n = i) = \pi_i$$

Regardless of the initial state, the probability of being in state i at time n approaches π_i as $n \to \infty$.

Renewal Interpretation

$$\pi_i = \frac{\text{expected sojourn time in } i}{\text{expected cycle length}}$$

5.5.3 Convergence Rates

For aperiodic chains, convergence to the stationary distribution is exponential:

$$|P_{ij}^{(n)} - \pi_j| \leq C\rho^n$$

where $0 < \rho < 1$ is the second-largest eigenvalue of the transition matrix, and C is a constant.

5.5.4 Applications in Practice

Simulation and MCMC

- Long-run proportions justify using single long runs instead of multiple independent runs - Burn-in periods allow the chain to approach stationarity - Sample averages approximate population means

Performance Analysis

- Server utilization in queueing systems - Long-run average costs in operational research - Steady-state analysis in engineering systems

Statistical Inference

- Estimating parameters from time series data - Testing for stationarity and ergodicity - Model validation through long-run behavior

5.5.5 Practical Considerations

- **Finite sample behavior**: Convergence may be slow for some chains
- **Mixing times**: Time required to reach approximate stationarity
- **Periodicity effects**: Periodic chains require Cesàro averaging
- **Initial distribution sensitivity**: Some chains reach stationarity faster than others

Chapter 6

Transient States & Absorption Analysis

The analysis of transient states and absorption processes forms a fundamental component of Markov chain theory. These concepts are essential for understanding systems that eventually reach terminal states, such as population extinction models, queueing systems with finite capacity, and various physical processes with absorbing boundaries.

6.1 Absorption Probabilities and Their Computation[19]

Absorption probabilities quantify the likelihood of reaching specific absorbing states from given transient states, providing crucial insights into long-term system behavior.

Definition 6.1 (Absorbing Markov Chain)**.** A Markov chain is called **absorbing** if:

1. It has at least one absorbing state (a state i where $p_{ii} = 1$)
2. From every non-absorbing state, it is possible to reach an absorbing state with positive probability

States that are not absorbing are called **transient states**.

For an absorbing Markov chain with t transient states and r absorbing states, the transition matrix can be written in canonical form:

$$P = \begin{pmatrix} Q & R \\ \mathbf{0} & I_r \end{pmatrix}$$

where:

- Q is a $t \times t$ matrix of transitions between transient states
- R is a $t \times r$ matrix of transitions from transient to absorbing states
- $\mathbf{0}$ is an $r \times t$ zero matrix
- I_r is the $r \times r$ identity matrix

6.1.1 Computing Absorption Probabilities

Let a_{ij} be the probability of eventual absorption in absorbing state j starting from transient state i. Using first-step analysis:

$$a_{ij} = \sum_k a_{kj} p_{ik} + r_{ij}$$

where the sum is over all transient states k, and r_{ij} is the probability of direct transition from state i to absorbing state j.

This system of linear equations can be solved iteratively or using matrix methods. The absorption probability vector $\mathbf{a}^{(j)}$ for absorbing state j satisfies:

$$\mathbf{a}^{(j)} = Q\mathbf{a}^{(j)} + \mathbf{r}^{(j)}$$

where $\mathbf{r}^{(j)}$ is the j-th column of R. Solving:

$$\mathbf{a}^{(j)} = (I - Q)^{-1}\mathbf{r}^{(j)}$$

6.2 First Passage Times and Mean Hitting Times[20]

First passage times characterize the temporal behavior of Markov chains, measuring how long it takes to reach specific states or sets of states.

Definition 6.2 (First Passage Time)**.** For a Markov chain $\{X_n\}$ and state j, the **first passage time** to state j is:

$$T_j = \inf\{n \geq 1 : X_n = j\}$$

The **mean hitting time** from state i to state j is $m_{ij} = \mathbb{E}_i[T_j]$.

6.2.1 Calculation Methods

First-Step Analysis

For mean hitting times, we can use the recursive relation:

$$m_{ij} = \begin{cases} 0 & \text{if } i = j \\ 1 + \sum_{k \neq j} p_{ik} m_{kj} & \text{if } i \neq j \end{cases}$$

This leads to the system $(I - Q_j)\mathbf{m}_j = \mathbf{1}$, where Q_j is the transition matrix with the j-th row and column removed.

Generating Function Approach

For more complex systems, generating functions provide powerful computational tools:

$$G_{ij}(s) = \mathbb{E}_i[s^{T_j}] = sp_{ij} + s\sum_{k \neq j} p_{ik} G_{kj}(s)$$

The mean hitting time is obtained as:

$$m_{ij} = \left.\frac{dG_{ij}(s)}{ds}\right|_{s=1}$$

Matrix Formulation

For absorbing chains, mean hitting times to the absorption set can be computed using:

$$\mathbf{m} = (I - Q)^{-1}\mathbf{1}$$

where $\mathbf{1}$ is the vector of ones.

6.3 Absorbing Markov Chains[21]

Absorbing Markov chains model processes that eventually terminate in one of several possible terminal states, making them invaluable for studying finite-horizon phenomena.

6.3.1 Fundamental Properties

Theorem 6.1 (Absorption Certainty)**.** In an absorbing Markov chain, absorption occurs with probability 1. That is, $\lim_{n\to\infty} Q^n = \mathbf{0}$.

Proof. From each transient state i, let m_i be the minimum number of steps to reach an absorbing state, and let $p_i < 1$ be the probability of not reaching an absorbing state in m_i steps. Let $m = \max_i m_i$ and $p = \max_i p_i < 1$.

The probability of not being absorbed in km steps is at most p^k, which approaches 0 as $k \to \infty$. Since non-absorption probabilities are monotone decreasing, $Q^n \to \mathbf{0}$. □

6.3.2 Powers of the Transition Matrix

For an absorbing chain in canonical form:

$$P^n = \begin{pmatrix} Q^n & (I - Q^n)NR \\ \mathbf{0} & I_r \end{pmatrix}$$

As $n \to \infty$:

$$\lim_{n\to\infty} P^n = \begin{pmatrix} \mathbf{0} & NR \\ \mathbf{0} & I_r \end{pmatrix}$$

The matrix NR gives the ultimate absorption probabilities.

6.4 Fundamental Matrix Approach[22]

The fundamental matrix provides a unified framework for analyzing absorbing Markov chains, enabling efficient computation of all relevant quantities.

Definition 6.3 (Fundamental Matrix)**.** For an absorbing Markov chain with transient states matrix Q, the **fundamental matrix** is:

$$N = (I - Q)^{-1} = I + Q + Q^2 + Q^3 + \cdots$$

The entry n_{ij} represents the expected number of times the process visits transient state j when starting from transient state i.

6.4.1 Key Properties and Applications

Expected Number of Steps to Absorption

The expected time to absorption starting from state i is:

$$t_i = \sum_j n_{ij}$$

In matrix form: $\mathbf{t} = N\mathbf{1}$

Absorption Probabilities

The probability of absorption in absorbing state j starting from transient state i is:

$$b_{ij} = \sum_k n_{ik} r_{kj}$$

In matrix form: $B = NR$

Variance of Absorption Times

The variance of absorption time from state i is:

$$\mathrm{Var}(T_i) = t_i(2N_{ii} - 1) - t_i^2$$

where N_{ii} is the i-th diagonal element of N.

6.4.2 Computational Aspects

The fundamental matrix can be computed using:

1. **Direct Inversion**: $N = (I - Q)^{-1}$
2. **Iterative Methods**: $N \approx \sum_{k=0}^{m} Q^k$ for sufficiently large m
3. **Recursive Formulas**: Using the relationship $N = I + QN$

6.5 Time-Reversibility and Detailed Balance[23]

Time-reversibility is a fundamental symmetry property in stochastic processes, crucial for understanding equilibrium behavior and developing efficient computational methods.

Definition 6.4 (Time-Reversible Markov Chain)**.** A stationary Markov chain is **time-reversible** if the reversed-time process has the same statistical properties as the forward-time process. Equivalently, the chain satisfies the **detailed balance conditions**:

$$\pi_i p_{ij} = \pi_j p_{ji} \quad \text{for all states } i, j$$

where π is the stationary distribution.

6.5.1 Physical Interpretation

Detailed balance states that in equilibrium, the probability flow from state i to state j equals the flow from j to i. This microscopic reversibility condition implies:

- No net probability current between any pair of states
- The process looks statistically identical when time is reversed
- Equilibrium represents true balance, not just steady-state flow

6.5.2 Applications in Statistical Mechanics

In statistical mechanics, detailed balance ensures that systems reach true thermal equilibrium. Key applications include:

Boltzmann Distribution

For a system with energy states E_i at temperature T:

$$\pi_i = \frac{e^{-E_i/(k_B T)}}{Z}$$

where Z is the partition function. Detailed balance requires:

$$\frac{w_{ij}}{w_{ji}} = e^{-(E_j - E_i)/(k_B T)}$$

where w_{ij} are transition rates.

Monte Carlo Methods

Detailed balance guides the construction of Markov Chain Monte Carlo (MCMC) algorithms:

- **Metropolis-Hastings**: Accept transitions with probability $\min(1, \frac{\pi_j q_{ji}}{\pi_i q_{ij}})$
- **Gibbs Sampling**: Update components using conditional distributions
- **Heat Bath Algorithm**: Direct sampling from equilibrium conditionals

6.6 Gambler's Ruin[24]

The gambler's ruin problem exemplifies absorbing Markov chains and demonstrates the practical application of fundamental matrix methods.

6.6.1 Problem Setup

A gambler starts with k dollars and plays against an opponent with $N - k$ dollars. At each round:

- The gambler wins \$1 with probability p
- The gambler loses \$1 with probability $q = 1 - p$
- The game ends when one player has \$0 (ruin) or \$N (victory)

The state space is $\{0, 1, 2, \ldots, N\}$ with absorbing states 0 and N.

6.6.2 Classical Analysis

Let P_k be the probability of eventual victory starting with k dollars. Using first-step analysis:

$$P_k = pP_{k+1} + qP_{k-1}, \quad k = 1, 2, \ldots, N-1$$

with boundary conditions $P_0 = 0$ and $P_N = 1$.

Solution for Fair Games ($p = q = 1/2$)

For fair games, the solution is:

$$P_k = \frac{k}{N}$$

The probability of ruin is $1 - P_k = \frac{N-k}{N}$.

Solution for Unfair Games ($p \neq q$)

For unfair games, let $r = q/p$. Then:

$$P_k = \begin{cases} \frac{1-r^k}{1-r^N} & \text{if } r \neq 1 \\ \frac{k}{N} & \text{if } r = 1 \end{cases}$$

6.6.3 Expected Duration

Let D_k be the expected number of rounds starting with k dollars:

$$D_k = 1 + pD_{k+1} + qD_{k-1}, \quad k = 1, 2, \ldots, N-1$$

with $D_0 = D_N = 0$.

Fair Game Duration

For fair games:

$$D_k = k(N-k)$$

Unfair Game Duration

For unfair games ($p \neq 1/2$):

$$D_k = \frac{k}{q-p} - \frac{N}{q-p} \cdot \frac{1-r^k}{1-r^N}$$

6.6.4 Fundamental Matrix Approach

For the gambler's ruin with $N = 4$, the transient states are $\{1, 2, 3\}$ with:

$$Q = \begin{pmatrix} 0 & p & 0 \\ q & 0 & p \\ 0 & q & 0 \end{pmatrix}$$

$$R = \begin{pmatrix} q & 0 \\ 0 & 0 \\ 0 & p \end{pmatrix}$$

The fundamental matrix $N = (I - Q)^{-1}$ provides:

- Expected visits to each transient state
- Expected game duration: $\mathbf{t} = N\mathbf{1}$
- Absorption probabilities: $B = NR$

6.6.5 Infinite Horizon Case

When the opponent has unlimited funds ($N \to \infty$):

- If $p \leq 1/2$: Ruin is certain ($P_k = 0$ for all k)
- If $p > 1/2$: Ruin probability is $(q/p)^k$

Expected time to ruin (when $p \leq 1/2$):

$$\mathbb{E}[T_{\text{ruin}}] = \frac{k}{q-p} \quad \text{if } p < 1/2$$

For $p = 1/2$, the expected time is infinite, though ruin occurs with probability 1.

6.6.6 Extensions and Applications

The gambler's ruin framework applies to numerous real-world scenarios:

- **Insurance**: Ruin probability for insurance companies
- **Inventory Management**: Stock-out analysis
- **Population Dynamics**: Extinction probabilities
- **Finance**: Portfolio optimization under constraints
- **Queueing Theory**: Finite buffer systems

Chapter 7

Modern Applications: PageRank Algorithms

7.1 Google's PageRank as Markov Chains[25][2]

Markov Chain Formulation

The PageRank algorithm models the web as a Markov chain. Consider the web as a directed graph $G = (V, E)$ with $|V| = n$ webpages. A directed edge $(j \to i)$ indicates that page j contains a hyperlink to page i.

We construct a column-stochastic matrix $M \in \mathbb{R}^{n \times n}$ from the hyperlink structure:

$$M_{ij} = \begin{cases} \frac{1}{d_j^{\text{out}}}, & \text{if page } j \text{ links to page } i, \\ 0, & \text{otherwise,} \end{cases}$$

where d_j^{out} denotes the number of outgoing hyperlinks on page j. Thus, the jth column of M describes the probability distribution of a random user *currently on page* j choosing one of its outgoing links uniformly at random.

Let $\mathbf{x}^{(k)} \in \mathbb{R}^n$ be a probability vector, where $\mathbf{x}_i^{(k)}$ is the probability that the user is on page i after k steps. A Markov chain evolves this vector as

$$\mathbf{x}^{(k+1)} = M\mathbf{x}^{(k)}, \quad \mathbf{x}^{(0)} = \frac{1}{n}\mathbf{1}.$$

Because M is column-stochastic (each column sums to 1), the vector remains normalized:

$$\|\mathbf{x}^{(k)}\|_1 = 1.$$

However, the raw link matrix M may fail to yield a unique stationary distribution because:

- Some pages may be **dangling nodes** (no outgoing links).
- The graph may not be strongly connected (**reducible**).
- The chain may have cycles (**periodicity**).

This motivates the introduction of teleportation.

Google Matrix and the Random Surfer Model

To guarantee aperiodicity and irreducibility, Google introduced teleportation through the *Google matrix*:

$$G = \beta M + (1-\beta)\frac{1}{n}\mathbf{1}\mathbf{1}^\top,$$

where $0 < \beta < 1$ is the *damping factor* (typically 0.85). Each column of G is a convex combination of:

- following hyperlinks from the current page (β weight),
- jumping uniformly to any page in the web ($1-\beta$ weight).

Operationally, the **random-surfer model** works as follows:

- With probability β: follow a random outgoing link.
- With probability $1-\beta$: teleport to a uniformly chosen page.

Because every page can be reached via teleportation, the Markov chain associated with G is now:

- **irreducible** — all states communicate,
- **aperiodic** — self-loops introduced by teleportation,
- **positive recurrent** — finite state space.

By the Perron–Frobenius theorem, G has a unique stationary distribution $\boldsymbol{\pi}$, called the **PageRank vector**. It satisfies:

$$\boldsymbol{\pi} = G^\top \boldsymbol{\pi}, \qquad \boldsymbol{\pi} \geq 0, \qquad \|\boldsymbol{\pi}\|_1 = 1.$$

Intuitively, π_i represents the long-run fraction of time a random surfer spends on page i.

Power Iteration

The PageRank vector is computed through the iterative scheme:

$$\mathbf{r}^{(k+1)} = G^\top \mathbf{r}^{(k)}, \qquad \mathbf{r}^{(0)} = \frac{1}{n}\mathbf{1}.$$

Since G is dense, storing it explicitly is infeasible for web-scale graphs. Instead, we exploit the decomposition of G:

$$G^\top \mathbf{r} = \beta M^\top \mathbf{r} + (1-\beta)\frac{\mathbf{1}^\top \mathbf{r}}{n}\mathbf{1}.$$

Because $\mathbf{r}$ is a probability vector, $\mathbf{1}^\top \mathbf{r} = 1$, so the formula simplifies to:

$$G^\top \mathbf{r} = \beta M^\top \mathbf{r} + \frac{1-\beta}{n}\mathbf{1}.$$

Thus, each iteration requires only:

1. one sparse matrix–vector multiplication with $M^\top$, and
2. adding a constant vector.

Convergence. The convergence of power iteration is linear:

$$\|\mathbf{r}^{(k)} - \boldsymbol{\pi}\|_1 \leq C\beta^k.$$

The rate is essentially governed by the damping factor β, since the second-largest eigenvalue magnitude of G is approximately β. In practice, 30–50 iterations yield high-accuracy PageRank scores.

7.2 Web Graph Structure and Challengess[2]

Structural Properties

Real web graphs are enormous, sparse, and exhibit heavy-tailed degree distributions. They are not strongly connected and contain many weakly connected components and dangling nodes (pages with no out-links).

Computational Challenges

Let $\mathrm{nnz}(M)$ denote the number of nonzeros in M. For web-scale graphs: - Storage must be sparse: $O(\mathrm{nnz}(M))$. - Each power-iteration step costs $O(\mathrm{nnz}(M))$. - Memory locality and I/O dominate performance.

Spectral Considerations

The dominant eigenvalue of G is 1, with the second eigenvalue bounded by $|\lambda_2(G)| \leq \beta$. The spectral gap $1 - |\lambda_2(G)|$ controls convergence speed.

7.3 Performance Evaluation and Modifications[26][27]

Acceleration Techniques

Extrapolation can reduce iterations. Let $\Delta\mathbf{r}^{(k)} = \mathbf{r}^{(k+1)} - \mathbf{r}^{(k)}$. One vector Aitken-style step is

$$\mathbf{r}^{(k+1)} \approx \mathbf{r}^{(k)} - \frac{\langle \Delta\mathbf{r}^{(k)}, \Delta^2\mathbf{r}^{(k)} \rangle}{\|\Delta^2\mathbf{r}^{(k)}\|_2^2}\, \Delta\mathbf{r}^{(k)},$$

where $\Delta^2\mathbf{r}^{(k)} = \Delta\mathbf{r}^{(k+1)} - \Delta\mathbf{r}^{(k)}$. Practical variants include Anderson acceleration and adaptive β.

Incremental and Streaming Updates

For dynamic graphs, warm-start from the previous $\mathbf{r}$ and update locally (e.g., push-based methods for Personalized PageRank). Monte Carlo approaches naturally support incremental updates via continued random-walk sampling.

Parallel and Distributed PageRank

- Power method parallelizes as sparse SpMV across threads/nodes; costs dominated by communication and partition quality. - Monte Carlo PageRank is embarrassingly parallel via independent random walks. - Block-structured/Page-block methods exploit host-level link locality to improve cache/I/O behavior.

Monte Carlo Estimation (Key Idea)

Run m random walks of average length n_{avg} with restart probability $1-\beta$. The visit frequency estimates $\boldsymbol{\pi}$. For node j with stationary mass π_j, the relative error satisfies approximately

$$\varepsilon \approx z_{1-\alpha/2}\sqrt{\frac{1-\pi_j}{\pi_j \cdot m n_{\mathrm{avg}}}},$$

making high-π_j nodes easier to estimate accurately.

Chapter 8

Conclusion

This report has presented a comprehensive analysis of Markov chains, from fundamental theoretical principles to practical applications. The elegant framework of stochastic processes governed by the memoryless property reveals how local transition dynamics determine global long-term behavior in remarkably predictable ways.

8.1 Key Findings

Markov chains are completely characterized by transition probability matrices, which encode system evolution. The classification of states—through concepts of communication, recurrence, transience, and periodicity—fundamentally determines chain behavior. Irreducibility and aperiodicity guarantee convergence to a unique stationary distribution, formalized through the ergodic theorem: for such chains, time averages converge almost surely to ensemble averages. Stationary distributions satisfy balance equations and represent long-run proportions in each state, with the relationship $\pi_i = 1/m_i$ connecting equilibrium probabilities to mean return times.

For absorbing chains, the fundamental matrix approach efficiently computes absorption probabilities, expected times to absorption, and variance—providing a unified computational framework. The PageRank application demonstrated how Markov chain theory elegantly addresses real-world problems at massive scale.

8.2 Method Selection Guidelines

Different problems require different approaches:

- **Small systems** (< 100 states): Direct linear system solution
- **Medium systems** (100–10,000 states): Power iteration method
- **Large sparse systems** (millions of states): Iterative methods with sparsity exploitation
- **Structured chains**: Analytical solutions exploiting special properties
- **Absorbing chains**: Fundamental matrix computation

8.3 Concluding Remarks

Markov chains provide both intellectual elegance and practical utility. The interplay between discrete matrix algebra and probabilistic evolution, between finite and infinite state spaces, and between deterministic long-run behavior and probabilistic short-term dynamics exemplifies the power of stochastic modeling. This foundational understanding enables analysis of complex systems across diverse fields, from web ranking to queuing networks to biological processes, making Markov chain theory an indispensable tool for the modern practitioner.

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